# Sepsis-3, severity scores and the Hour-1 bundle

## Definitions

- **Sepsis** = life-threatening organ dysfunction from a dysregulated host response to infection; **SOFA rise ≥ 2**
- **Septic shock** = vasopressor for MAP ≥ 65 **and** lactate > 2 mmol/L despite adequate fluid
- Sepsis-2: SIRS ≥ 2 + infection; 'severe sepsis' abolished; SIRS sensitive but non-specific

## Scores

- qSOFA (RR ≥ 22, altered mentation, SBP ≤ 100): prognostic only; SSC 2021 advises against it as sole screen
- NEWS2/MEWS: best for early deterioration; SOFA = diagnosis/trend; APACHE II (24 h) = mortality prediction

## Hour-1 bundle

- Lactate (repeat if > 2) · blood cultures ×2 · broad IV antibiotics < 1 h
- 30 mL/kg balanced crystalloid if hypotension or lactate ≥ 4 · noradrenaline if MAP < 65
- Look for source early (CT, ultrasound) and control it

**Key point:** Scores support, never replace, clinical judgement; time to antibiotics matters most in shock.

Notes: shock part 1 p.2

# Vasopressor escalation, steroids and beta-blockade

## Vasopressors

- Noradrenaline first line, peripheral start acceptable; target MAP ≥ 65 (60–65 acceptable if ≥ 65 yr: 65 trial)
- Add **vasopressin 0.03 U/min** when NA 0.25–0.5 µg/kg/min (VASST: benefit in less severe shock)
- Then adrenaline; dobutamine if low CO with myocardial dysfunction; angiotensin II (ATHOS-3) option

## Corticosteroids

- Hydrocortisone 200 mg/day if NA/adr ≥ 0.25 µg/kg/min ≥ 4 h (SSC 2021)
- ADRENAL: faster shock reversal, no mortality change; APROCCHSS (+ fludrocortisone): ↓ 90-day mortality

## Perfusion and beta-blockade

- Low ScvO2 + high lactate → inadequate DO2: check Hb, CO, fluid response
- Esmolol/landiolol for persistent tachycardia > 95 after resuscitation: Morelli 2013 positive, STRESS-L harm signal

**Key point:** Escalate in steps, review every hour: MAP, perfusion, lactate clearance, CRT.

# ROSE fluid model, VExUS and de-resuscitation

## ROSE phases

- **R**escue (bolus, minutes) → **O**ptimisation (titrate to responsiveness) → **S**tabilisation (maintain, zero balance) → **E**vacuation (de-resuscitate)
- Fluid is a drug: give only if responsive and still hypoperfused

## VExUS (venous congestion)

- Start with IVC > 2 cm; grade hepatic, portal and intrarenal venous Doppler
- Grade 0 none · 1 mild · 2 one severely abnormal · 3 ≥ 2 severely abnormal → predicts AKI

## De-resuscitation and fluid choice

- Diuretics or CRRT ultrafiltration; targets: negative balance, falling weight, EVLW, IAP
- Balanced crystalloids ≥ saline (SMART); albumin if large volumes (ALBIOS subgroup); avoid HES (6S, CHEST)
- CLASSIC, CLOVERS: restrictive ≈ liberal fluids

**Key point:** Positive balance after day 3 is linked with mortality: plan evacuation early.

# Oxygen delivery, ScvO2 and CO2 gap

## Oxygen delivery

- DO2 = CO × CaO2 × 10; CaO2 = 1.34 × Hb × SaO2 + 0.003 × PaO2 (kPa ×0.023)
- Normal DO2 ≈ 1000 mL/min (550–650 mL/min/m²); VO2 ≈ 250 mL/min; O2ER 20–30%

## Venous markers

- ScvO2 < 70%: low DO2 or high VO2; high ScvO2 (> 80%) may mean shunting/impaired extraction
- Pcv–aCO2 gap > 6 mmHg = low flow even if ScvO2 normal; gap/C(a–v)O2 ratio > 1.4 = anaerobic metabolism
- Lactate: type A (hypoperfusion) vs type B (adrenaline, liver failure, metformin)

## Improving tissue oxygenation

- Raise CO (fluid if responsive, dobutamine), Hb if < 7 (or ischaemia), SaO2
- Reduce VO2: sedation, ventilation, cooling fever, control shivering

**Key point:** Normal ScvO2 does not exclude hypoperfusion: always add the CO2 gap and lactate.

Notes: shock part 1 p.1

# Shock classification and ultrasound approach

## Definition and types

- Shock = circulatory failure causing inadequate cellular O2 use; lactate > 2 mmol/L
- Hypovolaemic and cardiogenic: low CO, high SVR; obstructive: low CO, high CVP; distributive: high CO, low SVR

## Ultrasound (RUSH / pump-tank-pipes)

- Pump: LV function, RV dilatation, effusion/tamponade
- Tank: IVC size/collapse, FAST, pleural fluid, B-lines (lung oedema)
- Pipes: aortic aneurysm/dissection, DVT

## Resuscitation targets

- MAP ≥ 65, CRT normal (ANDROMEDA-SHOCK), UO > 0.5 mL/kg/h, lactate falling
- Treatment follows type: fluids/blood, inotrope, decompression/drainage/lysis, vasopressor + source control

**Key point:** Classify early with ultrasound; the shock type decides the treatment.

Notes: shock part 1 p.22

# Anaphylactic shock

## Immediate management

- Stop trigger, call help, lie flat/legs up, high-flow O2
- **IM adrenaline 0.5 mg (1:1000) anterolateral thigh**, repeat every 5 min
- Fluid 500–1000 mL bolus; refractory: IV adrenaline infusion 0.05–0.1 µg/kg/min
- Airway oedema: early intubation by expert; nebulised adrenaline as a bridge

## Beta-blocked patient

- Bisoprolol blunts adrenaline response → refractory hypotension, bradycardia
- **Glucagon 1–5 mg IV over 5 min**, then 5–15 µg/min infusion; vasopressin/NA if needed

## Confirmation and follow-up

- Mast-cell tryptase at ≈ 1 h and 2–4 h, baseline at 24 h
- Antihistamines/steroids are second-line only; observe 6–12 h (biphasic risk)
- Allergy referral, adrenaline auto-injector, report the reaction

**Key point:** Adrenaline IM is the only first-line drug; never delay it for antihistamines.

# Vitamin C, adjuncts and post-sepsis syndrome

## Vitamin C evidence

- Marik 2017 (HAT, before–after) positive; VITAMINS, ACTS, VICTAS neutral
- **LOVIT 2022: high-dose vitamin C ↑ death or persistent organ dysfunction** → not recommended (SSC 2021)

## Other adjuncts

- Hydrocortisone in refractory shock: yes · IVIG, selenium, statins, polymyxin B haemoperfusion: no
- Bicarbonate only if pH ≤ 7.2 + AKI (BICAR-ICU); blood purification: research only
- Insulin target 8–10 mmol/L (NICE-SUGAR)

## Post-sepsis syndrome

- Physical weakness, cognitive impairment, PTSD/depression, recurrent infection, readmission
- Follow-up clinic, rehabilitation, medication review, family support, ICU diaries

**Key point:** Most sepsis 'magic bullets' failed; good basic care and follow-up improve outcome.

# Fever versus hyperthermia in ICU

## Definitions

- **Fever**: raised hypothalamic set point (cytokines, PGE2), responds to antipyretics
- **Hyperthermia**: set point normal, heat production/loss failure (NMS, MH, serotonin, heat stroke); antipyretics useless

## Causes of fever in ICU

- Infective: VAP, CRBSI, UTI, sinusitis, C. difficile, surgical site, abscess
- Non-infective: drugs, DVT/PE, pancreatitis, transfusion, acalculous cholecystitis, adrenal crisis, thyroid storm, CNS bleed

## Work-up and treatment

- Exam, lines, cultures (blood ×2, sample site), CXR, imaging of source; PCT trend
- Paracetamol for comfort; HEAT/REACTOR: no mortality benefit from treating fever
- Active cooling for hyperthermia and after brain injury/cardiac arrest (avoid > 37.7)

**Key point:** Fever > 41 °C or no response to antipyretics → think hyperthermia syndrome.

# Antibiotic PK/PD in critical illness

## PK changes

- ↑ Volume of distribution (capillary leak, fluids, ECMO) → higher loading dose
- Augmented renal clearance (CrCl > 130) → underdosing; AKI/CRRT alter clearance
- Low albumin ↑ free fraction of highly bound drugs (ceftriaxone, teicoplanin)

## PK/PD targets

- Time-dependent (beta-lactams): fT > MIC → **extended/continuous infusion** (BLING III, MERINO)
- Concentration-dependent (aminoglycosides): Cmax/MIC 8–10 → once daily
- AUC/MIC: vancomycin (AUC 400–600), quinolones

## TDM and stewardship

- TDM for vancomycin, aminoglycosides, and beta-lactams where available (TARGET)
- Full loading dose regardless of renal function; adjust maintenance only

**Key point:** Early full loading + prolonged beta-lactam infusion = right drug, right dose.

# Antimicrobial stewardship and procalcitonin

## Stewardship principles

- Right drug, dose, route, timing, duration; cultures before antibiotics
- Daily review: 48–72 h 'time-out', de-escalate by culture, switch IV → oral
- Local antibiogram, restricted lists, ID/microbiology input, audit and feedback

## Procalcitonin

- Stop when PCT < 0.5 µg/L or ≥ 80% fall from peak (SAPS: ↓ duration and mortality, PRORATA)
- Not for starting antibiotics in shock; false rise in surgery, trauma, burns, pancreatitis

## Duration

- 7 days for most (VAP, intra-abdominal after source control: STOP-IT 4 days)
- Uncomplicated Gram-negative bacteraemia 7 days (BALANCE: 7 = 14)
- Longer: S. aureus bacteraemia, endocarditis, undrained abscess, osteomyelitis

**Key point:** Shorter courses are as good and reduce resistance and C. difficile.

# Multidrug-resistant Gram-negative infection

## Risk factors

- Prior antibiotics within 90 days, long ICU stay, MV, devices, previous colonisation
- Travel/hospital in high-prevalence area, dialysis, nursing home, immunosuppression

## Empirical choice (local ecology)

- ESBL: carbapenem (MERINO: meropenem > pip-taz)
- CRE (KPC/OXA-48): ceftazidime-avibactam; MBL (NDM): ceftazidime-avibactam + aztreonam
- CRAB: colistin or sulbactam-durlobactam combinations; CR Pseudomonas: ceftolozane-tazobactam
- Colistin: loading 9 MU, nephrotoxic; combine and de-escalate once sensitivities known

## Infection control

- Contact precautions, single room/cohorting, hand hygiene, screening swabs
- Chlorhexidine bathing, environmental cleaning, stewardship

**Key point:** Know your antibiogram: the empirical regimen depends on local resistance.

# Catheter-related bloodstream infection

## Definitions

- **CLABSI**: surveillance, BSI with CVC > 48 h, no other source
- **CRBSI**: clinical, catheter proven as source: DTP ≥ 2 h (catheter culture positive ≥ 2 h earlier) or same organism on tip (≥ 15 CFU)

## Management

- Remove line if: S. aureus, Candida, Pseudomonas, tunnel/pocket infection, shock, persistent bacteraemia
- Empirical vancomycin ± Gram-negative cover; duration 7–14 days; S. aureus ≥ 14 days + echo
- Lock therapy only for long-term lines with CoNS when removal impossible

## Prevention bundle

- Hand hygiene, maximal barrier, **chlorhexidine 2% in alcohol**, subclavian preferred, ultrasound
- Daily review, remove unneeded lines, chlorhexidine dressings

**Key point:** Persistent bacteraemia after 72 h → think endocarditis or septic thrombosis.

# Peritonitis, source control and abdominal hypertension

## Classification

- Primary (SBP), secondary (perforation/leak), tertiary (persistent after adequate treatment)
- Tertiary: immunosuppression, resistant flora (Enterococcus, Candida, CoNS), high mortality

## Source control

- Drain, debride, divert, restore: within 6–12 h; percutaneous drainage if possible
- Damage control laparotomy if unstable; relook if needed

## Intra-abdominal hypertension

- IAH ≥ 12 mmHg; ACS > 20 + new organ failure; APP = MAP − IAP (target ≥ 60)
- Decompress (NGT, rectal tube), drain ascites, sedation/NMB, negative balance, laparostomy

## Antibiotics

- Pip-taz or carbapenem; add antifungal and anti-enterococcal cover in tertiary peritonitis
- 4 days after source control (STOP-IT)

**Key point:** Antibiotics fail without source control: drain early.

# Invasive candidiasis

## Risk factors

- Broad antibiotics, CVC, TPN, abdominal surgery/leak, pancreatitis, dialysis
- Multifocal colonisation (Candida score ≥ 3), immunosuppression, long ICU stay

## Diagnosis

- Blood cultures ≈ 50% sensitive; beta-D-glucan (high NPV, false positives: albumin, IVIG, dialysis)
- Mannan/anti-mannan, T2Candida PCR; eye exam and echo if candidaemia

## Treatment

- **Echinocandin first line** (caspofungin, anidulafungin, micafungin)
- Step-down to fluconazole if sensitive and stable; remove CVC
- 14 days after first negative culture and resolution

## Strategy

- Prophylaxis: high-risk without infection (e.g. recurrent leak)
- Pre-emptive: biomarker/colonisation guided; empirical: septic shock with risk factors

**Key point:** Candidaemia is never a contaminant: treat, remove the line, look for endophthalmitis.

# Invasive pulmonary aspergillosis

## Risk factors

- Neutropenia, haematological malignancy, HSCT, solid organ transplant, high-dose steroids
- ICU: COPD on steroids, cirrhosis, severe influenza (IAPA) or COVID-19 (CAPA)

## Diagnosis

- CT: nodules, halo sign (early), air-crescent (late); wedge infarcts
- BAL galactomannan (> 1.0), serum galactomannan (neutropenic), Aspergillus PCR, culture
- EORTC/MSG (host + clinical + mycology); AspICU for non-neutropenic ICU patients

## Treatment

- **Voriconazole** (or isavuconazole) first line; liposomal amphotericin if azole resistant
- TDM for voriconazole (1–5.5 mg/L); visual disturbance, hepatotoxicity, QT
- Interactions: CYP3A4 inhibition (tacrolimus, ciclosporin); 6–12 weeks

**Key point:** Influenza patients deteriorating in ICU: send BAL galactomannan.

# Pyelonephritis, urosepsis and CAUTI

## Diagnosis

- Fever, loin pain, pyuria; complicated if obstruction, male, catheter, pregnancy, diabetes
- Emphysematous pyelonephritis: gas on CT, diabetics (Huang–Tseng classes 1–4)

## Management

- Cultures, broad antibiotics (cephalosporin/pip-taz; carbapenem if ESBL risk)
- **Imaging for obstruction → urgent decompression** (nephrostomy or stent)
- Emphysematous: drainage, nephrectomy for extensive disease

## CAUTI prevention

- Avoid unnecessary catheters; remove early (nurse-led protocols)
- Aseptic insertion, closed drainage, bag below bladder; treat only symptomatic bacteriuria

**Key point:** Obstructed infected kidney is a urological emergency: drain it.

# Diabetic foot infection

## Severity (IDSA/IWGDF)

- Mild: local skin; moderate: deeper or > 2 cm erythema; severe: SIRS/systemic signs
- Assess pulses, ABPI, probe-to-bone (osteomyelitis), X-ray/MRI

## Microbiology

- Polymicrobial: S. aureus, streptococci, Gram-negatives, anaerobes; MRSA/Pseudomonas risks

## Management

- Glycaemic control, fluids, broad antibiotics (pip-taz ± vancomycin)
- **Urgent surgical debridement** of abscess, gas, necrosis; amputation if needed
- Vascular assessment and revascularisation; offloading; wound care

## Poor outcome predictors

- PAD/ischaemia, sepsis, osteomyelitis, high HbA1c, CKD, wet gangrene, delay

**Key point:** Severe diabetic foot = sepsis + surgery: debride early.

# Febrile neutropenia

## Definition and first hour

- Neutrophils < 0.5 ×10^9/L (or falling) + temperature ≥ 38.3 once or ≥ 38 for 1 h
- Cultures, then anti-pseudomonal beta-lactam **within 1 h** (pip-taz, meropenem if ESBL)
- Add vancomycin for line infection, MRSA, skin/soft tissue, shock

## Antifungals

- Persistent fever after 4–7 days of antibiotics → CT chest, galactomannan
- Empirical/pre-emptive echinocandin or liposomal amphotericin; voriconazole for mould

## Complications

- Neutropenic enterocolitis (typhlitis): CT, bowel rest, cover anaerobes, surgery if perforation
- Tumour lysis, mucositis, Pneumocystis, CMV

## G-CSF and outcome

- G-CSF not routine for established FN; consider in high-risk/sepsis
- Mortality ↑ with shock, MV, invasive fungal infection

**Key point:** Antibiotics within 60 minutes save lives in neutropenic sepsis.

# Infective endocarditis

## Modified Duke criteria

- Major: typical organism in 2 cultures, echo evidence (vegetation, abscess, new regurgitation)
- Minor: predisposition, fever ≥ 38, vascular, immunological phenomena, atypical microbiology
- Definite: 2 major, 1 major + 3 minor, or 5 minor

## Antibiotics

- Empirical: vancomycin + gentamicin (native) or + rifampicin (prosthetic)
- MSSA: flucloxacillin/cefazolin; streptococci: penicillin/ceftriaxone; 4–6 weeks

## Urgent surgery

- Heart failure from valve dysfunction (emergency), uncontrolled infection/abscess
- Embolism prevention: vegetation > 10 mm with embolic event

## ICU complications

- Heart failure, emboli (stroke, spleen, kidney), mycotic aneurysm, AKI, heart block

**Key point:** New heart block in endocarditis = root abscess until proven otherwise.

# Necrotising soft-tissue infection

## Classification

- Type I polymicrobial (diabetes, perineum: Fournier's); type II group A strep ± S. aureus
- Type III Clostridium/Vibrio; type IV fungal

## Diagnosis

- Pain out of proportion, rapid spread, bullae, crepitus, shock
- LRINEC ≥ 6 suggests (poor sensitivity); CT gas/fascial fluid; **surgery is diagnostic**

## Management

- **Urgent wide debridement**, relook in 24 h
- Pip-taz/carbapenem + **clindamycin** (toxin suppression) ± vancomycin/linezolid
- IVIG for streptococcal TSS (considered); HBO not a delay

## Streptococcal toxic shock

- Hypotension + ≥ 2 organ failures with group A strep; rash, desquamation
- Source control, clindamycin, supportive care

**Key point:** Never wait for imaging when NSTI is suspected clinically.

# Severe malaria

## Severity features (P. falciparum)

- Impaired consciousness, seizures, shock, ARDS, AKI, acidosis, hypoglycaemia
- Parasitaemia > 10%, Hb < 7, jaundice, bleeding, haemoglobinuria

## Specific treatment

- **IV artesunate 2.4 mg/kg at 0, 12, 24 h**, then daily (AQUAMAT, SEAQUAMAT)
- Switch to oral ACT after ≥ 24 h when able; quinine only if no artesunate
- Watch delayed haemolysis 1–3 weeks later

## Supportive care

- Cautious fluids (FEAST: boluses harmful in children), glucose monitoring
- Treat seizures, RRT early, ventilation, transfusion; antibiotics for co-infection
- No steroids, mannitol or heparin; exchange transfusion not recommended

**Key point:** Artesunate beats quinine: start it without delay.

# Dengue

## Classification (WHO 2009)

- Dengue without/with warning signs; severe dengue (plasma leak shock, bleeding, organ failure)
- Warning signs: abdominal pain, vomiting, fluid accumulation, mucosal bleed, lethargy, liver ↑, rising HCT + falling platelets

## Phases

- Febrile (2–7 d) → **critical (defervescence, 24–48 h leak)** → recovery (reabsorption)

## Fluids in shock

- Crystalloid 10–20 mL/kg boluses guided by HCT and perfusion; taper over 24–48 h
- Colloid if refractory; rising HCT = more leak, falling HCT + instability = bleeding → blood
- Stop fluids in recovery phase (overload risk)

## Platelets

- Transfuse only for active bleeding or procedures; not prophylactic

**Key point:** The danger comes when the fever falls: watch the critical phase closely.

# Leptospirosis

## Diagnosis

- Exposure to water/rats (floods, farmers); fever, myalgia (calves), conjunctival suffusion
- **Weil's disease**: jaundice + AKI + bleeding; pulmonary haemorrhage = worst
- Tests: PCR (first week), IgM ELISA, MAT; low K AKI (non-oliguric)

## Management

- **Penicillin G or ceftriaxone** IV; doxycycline for mild
- Supportive: early RRT, lung-protective ventilation, ECMO for haemorrhage
- Jarisch–Herxheimer reaction possible

## Differentials

- Malaria, dengue, hantavirus, typhoid, viral hepatitis, rickettsia, sepsis

**Key point:** Fever + jaundice + AKI after flood exposure = leptospirosis.

# Strongyloides hyperinfection

## Pathophysiology

- Autoinfection cycle; steroids/HTLV-1/transplant → hyperinfection and dissemination
- Larvae carry gut bacteria → Gram-negative bacteraemia and meningitis

## Diagnosis

- Eosinophilia (often absent in hyperinfection), larvae in stool, sputum or BAL
- Serology (IgG), agar plate culture, PCR

## Treatment and prevention

- **Ivermectin 200 µg/kg daily** until negative for 2 weeks; SC/rectal if no gut absorption
- Treat Gram-negative sepsis; reduce immunosuppression
- Screen/treat before steroids in endemic patients (e.g. Nile Delta)

**Key point:** Gram-negative sepsis on steroids from an endemic area → check for Strongyloides.